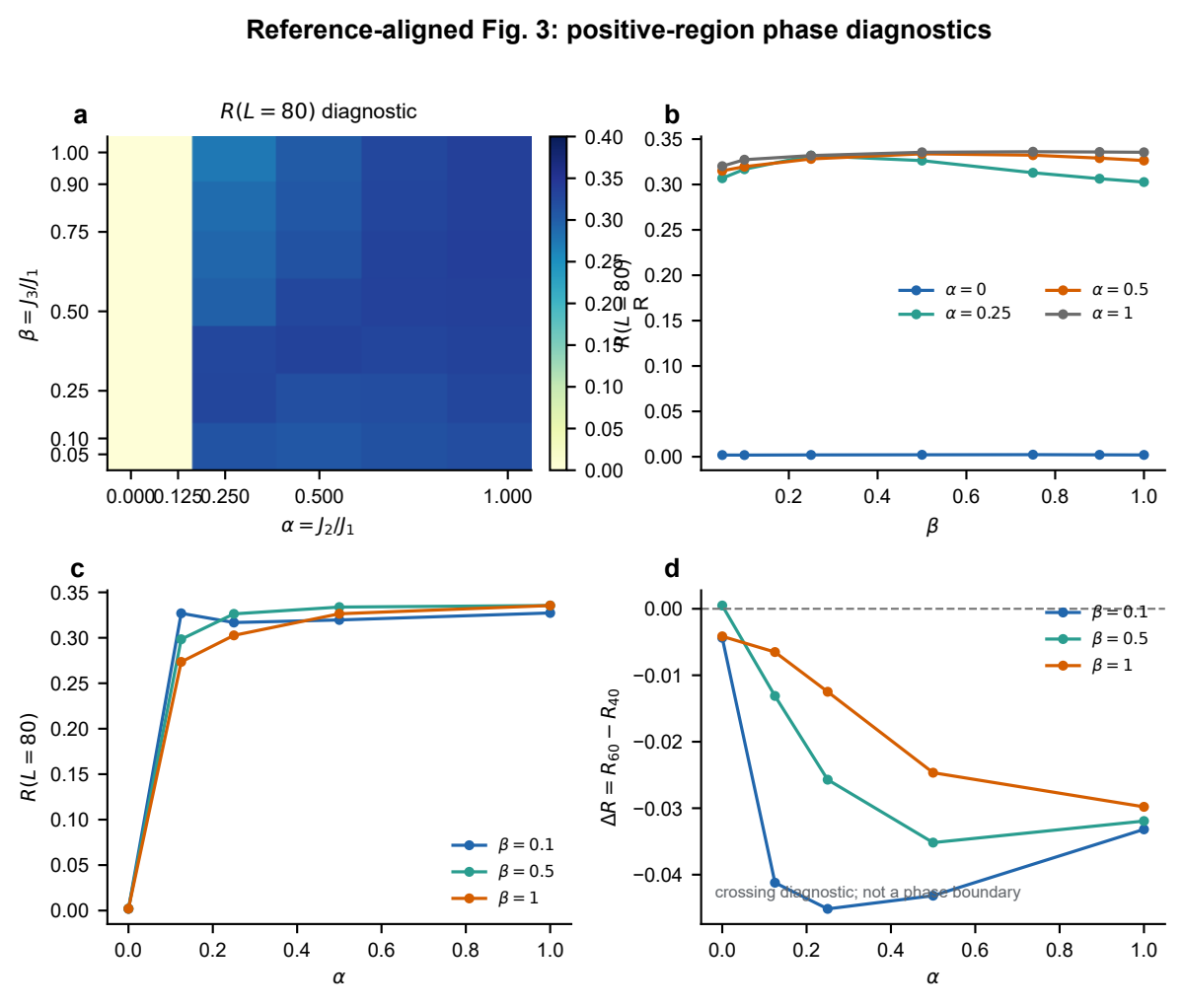

Reference-aligned Fig. 3: positive-region phase diagnostics
a
$R(L=80)$ diagnostic
$\beta = J_3/J_1$
$\alpha = J_2/J_1$
b
$R(L=80)$
β
$\alpha = 0$
$\alpha = 0.25$
$\alpha = 0.5$
$\alpha = 1$
c
$R(L=80)$
α
$\beta = 0.1$
$\beta = 0.5$
$\beta = 1$
d
$\Delta R = R_{60} - R_{40}$
α
$\beta = 0.1$
$\beta = 0.5$
$\beta = 1$
crossing diagnostic; not a phase boundary